MILITARY SPECIFICATION

FUZE, HAND GRENADE, M227

This specification is approved for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 This specification covers one type of fuze.

2. APPLICABLE DOCUMENTS

2.1 Government documents. The following documents of the issue in effect on date of invitation for bids or request for proposal, form a part of this specification to the extent specified herein:

SPECIFICATIONS

FEDERAL

PPP-B-636 - Box, Fiberboard.
PPP-F-320 - Fiberboard; Corrugated and Solids, Sheet Stock (Container Grade) and Cut Shapes.

MILITARY

MIL-B-117 - Bags and Sleeves and Tubing, Interior Packaging.
MIL-B-131 - Barrier Materials Water Vapor Proof, Flexible, Heat Sealable.

STANDARDS

MILITARY

MIL-STD-105 - Sampling Procedures and Tables for Inspection by Attributes.
MIL-STD-129 - Marking for Shipment and Storage.
MIL-STD-331 - Fuze and Fuze Components, Environmental and Performance Test for.
MIL-STD-810 - Environmental Test Methods.
MIL-STD-1168 - Lot Numbering of Ammunition.

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DRAWINGS

US ARMY ARMAMENT COMMAND

EDGEWOOD ARSENAL

D13-10-40 - Fuze, Hand Grenade, M227.

(Copies of specifications, standards, drawings and publications required by suppliers in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

3. REQUIREMENTS

3.1 Materials and components.

3.1.1 Materials. All materials cited on Drawing D13-10-40 or on the subsidiary drawings shall conform to the specifications listed thereon, or to the specific characteristics set forth on the drawings.

3.1.2 Components. All components of the fuze shall conform to the specifications and drawings listed on Drawing D13-10-40 and on subsidiary drawings.

3.2 Assembly. The fuze shall be assembled as shown on Drawing D13-10-40.

3.3 Functioning. The fuze shall exhibit the following characteristics when tested in accordance with 4.4.4.1:

(a) The safety pin shall be extracted when a force of 10 to 30 pounds is applied along the axis of the pin.
(b) The safety latch shall move from full engagement to full disengagement when a force of 1.5 to 4.0 pounds is applied tangentially to the latch.
(c) After release of the handle, the pyrotechnic mix shall propagate and emit burning particles from the end of the fuze in 0.1 to 0.9 seconds.

3.4 Environmental.

3.4.1 Jolt. The fuze shall comply with MIL-STD-331 after testing as specified in 4.3.4.1.

3.4.2 Jumble. The fuze shall comply with MIL-STD-331 after testing as specified in 4.3.4.2.

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3.4.3 Five-foot drop. The fuze shall comply with the requirements of MIL-STD-331 and 3.3 after testing when tested as specified in 4.3.4.3.

3.4.4 Vibration. The fuze shall comply with 3.3 after testing as specified in 4.3.4.4.

3.4.5 Leakage. The fuze shall comply with MIL-STD-810 and 3.3 after testing as specified in 4.4.4.1.

3.5 Preproduction. Prior to the start of regular production, a preproduction lot of fuzes shall be produced in accordance with this specification for examination and test.

3.6 Workmanship. The fuze shall be free from foreign matter (excessive thread sealant, oil, grease or dirt) and damage such as cracks, dents or deformation which could affect its intended use.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection.

4.1.1 Supplier's responsibility. Unless otherwise specified in the contract or purchase order, the supplier is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified in the contract or order, the supplier may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to specified requirements.

4.1.2 Objective evidence. The supplier shall provide objective evidence acceptable to the contracting officer that the requirements of 3.1 and section 5 for which specific inspection has not been provided in this specification have been satisfied.

4.2 Classification of inspections.

(a) Preproduction inspection (see 4.3).
(b) Quality conformance inspection (see 4.4).

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4.3 Preproduction inspection.

4.3.1 Lot. A preproduction lot of 500 fuzes shall be manufactured using the same methods, materials, equipment and processes as will be used during regular production.

4.3.2 Sampling.

4.3.2.1 For examination. Sampling shall be conducted in accordance with MIL-STD-105.

4.3.2.2 For destructive tests.

(a) Functioning. Sampling shall be conducted in accordance with MIL-STD-105.
(b) Jolt. Sampling shall be conducted in accordance with MIL-STD-105, level S-4.
(c) Jumble. Sampling shall be conducted in accordance with MIL-STD-105, level S-4.

4.3.3 Inspection procedure.

4.3.3.1 For examination. The sample fuzes shall be examined in accordance with the classification of defects and with MIL-STD-105. The AQLs do not apply.

4.3.3.2 For destructive tests.

(a) Functioning. The sample fuzes shall be tested in accordance with 4.3.4.3, 4.3.4.4, and 4.4.4.1 in that order and with table I.
(b) Jolt. The sample fuzes shall be tested in accordance with 4.3.4.1 using an AQL of 1.5.
(c) Jumble. The sample fuzes shall be tested in accordance with 4.3.4.2 using an AQL of 1.5.

4.3.4 Tests. Tests shall be conducted as follows:

4.3.4.1 Jolt. Test the fuze in accordance with test 101 of MIL-STD-331.

4.3.4.2 Jumble. Test the fuze in accordance with test 102 of MIL-STD-331.

4.3.4.3 Five-foot drop. Test the fuze in accordance with test 111 of MIL-STD-331, except that breakdown inspection is not required.

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4.3.4.4 Vibration. Test the fuze in accordance with test 104, procedure II of MIL-STD-331, except that assembly to inert grenade and breakdown inspection is not required.

4.3.5 Acceptance/rejection criteria. The preproduction lot shall comply with the examinations and tests specified in 4.3.3 to be acceptable. The supplier shall obtain written approval from the contracting officer prior to proceeding with regular production.

4.4 Quality conformance inspection.

4.4.1 Lotting. Each lot shall be identified and controlled in accordance with MIL-STD-1168. However, no more than one interfix number of each component (primers, first fire pellets, delay pellets and ignition mix) from one manufacturer shall be contained in any one lot of assembled fuzes.

4.4.2 Sampling.

4.4.2.1 For examination. Sampling shall be conducted in accordance with MIL-STD-105.

4.4.2.2 For destructive tests. Sampling shall be conducted in accordance with MIL-STD-105, inspection level S-4.

4.4.3 Inspection procedure.

4.4.3.1 For examination. The sample fuzes and the level C preparation for delivery shall be examined in accordance with the classification of defects and with MIL-STD-105.

4.4.3.2 For destructive tests. The sample fuzes shall be tested in accordance with 4.4.4.1 and with table I.

Table I. Functioning

Characteristic	AQL (percent defective)
Safety pin pull (see 3.3(a))	2.5
Safety latch force (see 3.3(b))	
Functioning time (see 3.3(c))	0.40

4.4.3.3 Critical defects. Each fuze in the lot shall be inspected for its critical characteristics listed in the classification of defects.

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4.4.3.4 Classification of defects.

(a) Fuze, hand grenade, M227 (Drawing D13-10-40).

Categories	Defects
Critical:	
1	Safety pin missing
2	Safety latch missing
3	Safety latch spring missing or incorrect
Major: AQL 1.0 percent defective	
101	Component missing or incorrectly assembled
102	Shellac missing
103	Marking missing or incorrect
104	Thread incorrect
105	Protective finish missing or incorrectly applied
106	Cement missing
107	Workmanship (3.6)

(b) Preparation for delivery (section 5).

Categories	Defects
Critical:	None defined
Major: AQL 2.5 percent defective	
101	Marking missing or incorrect
102	Container closure incorrect
103	Partitions incorrect
104	Container incorrect
105	Barrier bag incorrect
106	Barrier bag incorrectly sealed or ruptured
107	Packing container incorrect

4.4.4 Tests.

4.4.4.1 Leakage and functioning.

(a) The fuze shall be tested as specified in Method 512, procedure I of MIL-STD-810, except that the last sentence of paragraph 3.1.3

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does not apply. The fuze shall be functioned within 1 hour after completing the test.

(b) Using suitable test equipment, place fuze in vertical position, arm and function fuze to determine:

(1) The force required to extract the safety pin.
(2) The force required to move the safety latch.
(3) The time interval from release of the handle to emission of burning particles.

5. PREPARATION FOR DELIVERY

5.1 Packaging, level C. One hundred fuzes shall be packaged in a fiberboard box conforming to style RSC grade W5c of PPP-B-636 and having inside dimensions of 21 1/4 inches long by 10 5/8 inches wide by 6 1/4 inches deep. The box shall be fitted with fiberboard partitions forming two tiers of 50 cells each in 5 rows of 10 cells per row. Each cell shall measure 2 inches by 2 inches by 3 inches (length, width, and depth respectively). The two tiers shall be separated by a fiberboard separator conforming to grade W5c of PPP-F-320 and measuring 21 1/8 inches long by 10 1/2 inches wide. Each fuze shall be placed in an upright position within a cell and the fiberboard box shall be closed in accordance with the appendix to PPP-B-636 using Method IV. The fiberboard box shall then be placed within a barrier bag measuring 34 inches long by 18 inches wide (inside useable dimensions) conforming to MIL-B-117, type I, Class E, style 1, the material of which is backed by either spunbonded polyolefin or spunbonded nylon material, per MIL-B-131. The barrier bag shall be closed by heat sealing and the included air volume shall be kept to a practicable minimum.

5.2 Packing, level C. Two boxes of fuzes packaged as specified in 5.1 shall be packed together in a fiberboard box conforming to PPP-B-636, style RSC, Grade W5c having inside dimensions of 21 inches long by 11 1/4 inches wide by 13 3/4 inches deep. The box shall then be closed in accordance with the appendix to PPP-B-636 using Method V closure requirements.

5.3 Marking. Unless otherwise specified in the contract or purchase order, unit packages, barrier bags and the shipping container shall be marked in accordance with the domestic shipping requirements of MIL-STD-129 to include as a minimum, the following:

National Stock Number
Item Nomenclature
Quantity and Unit of Issue
Date packed Level of Protection
Contractors Name and Contract Number
Lot Number

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6. NOTES

6.1 Intended use. This item is intended for use as an ignition element for the M47 and M48 grenades.

6.2 Ordering data. Procurement documents should specify the following:

(a) Title, number and date of this specification.
(b) Preproduction.

(1) Time allowed for supplier submission of samples for Government test and evaluation after award of contract.
(2) Name and address of test facility and shipping instructions when testing is performed by the Government.
(3) Time required for the Government to notify the supplier whether to proceed with production.

6.3 Caution. Suppliers are hereby notified to take all necessary precautions to protect personnel during mixing of pyrotechnic materials, palletizing and loading of fuzes to prevent fire, explosion and personal injury.

6.4 Supersession data. This specification supersedes Edgewood Arsenal Purchase Description EA-F-1013C dated 28 August 1972.

Custodian:

Army - EA

Preparing activity:

Army - EA

Project No. 1330-A109

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SPECIFICATION ANALYSIS SHEET

Form Approved
Budget Bureau No. 22-R255

INSTRUCTIONS: This sheet is to be filled out by personnel, either Government or contractor, involved in the use of the specification in procurement of products for ultimate use by the Department of Defense. This sheet is provided for obtaining information on the use of this specification which will insure that suitable products can be procured with a minimum amount of delay and at the least cost. Comments and the return of this form will be appreciated. Fold on lines on reverse side, staple in corner, and send to preparing activity. Comments and suggestions *submitted on this form* do not constitute or imply authorization to waive any portion of the referenced document(s) or serve to amend contractual requirements.

SPECIFICATION

ORGANIZATION

CITY AND STATE

CONTRACT NUMBER

MATERIAL PROCURED UNDER A

☐ DIRECT GOVERNMENT CONTRACT ☐ SUBCONTRACT

1. HAS ANY PART OF THE SPECIFICATION CREATED PROBLEMS OR REQUIRED INTERPRETATION IN PROCUREMENT USE?

   A. GIVE PARAGRAPH NUMBER AND WORDING.

   B. RECOMMENDATIONS FOR CORRECTING THE DEFICIENCIES

2. COMMENTS ON ANY SPECIFICATION REQUIREMENT CONSIDERED TOO RIGID

3. IS THE SPECIFICATION RESTRICTIVE?

   ☐ YES ☐ NO *(If "yes", in what way?)*

4. REMARKS *(Attach any pertinent data which may be of use in improving this specification. If there are additional papers, attach to form and place both in an envelope addressed to preparing activity)*

SUBMITTED BY *(Printed or typed name and activity - Optional)*

DATE

DD FORM 1 JAN 66 1426 REPLACES EDITION OF 1 OCT 64 WHICH MAY BE USED.

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OFFICIAL BUSINESS

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